

CUI

Federal Agency: Army

Program Office: Army Contracting Command-Red River Army Depot (ACC-RRAD)

Aol Title: Track Pin and Bushing Manufacturing

CSO Aol Number: 004

Aol Release Date: 30 Oct 2025

DISTRIBUTION STATEMENT D: Distribution authorized to the Department of Defense and U.S. DoD contractors or grantees, Unclassified Technical Information, 08 October 2025. Other request for this document must be referred to ACC-RRAD.

Controlled By: U.S. ARMY Controlled By: ACC-RRAD CUI Category(ies): CTI//OPSEC Distribution Statement: D POC: usarmy.rrad.acc.mbx.cso@army.mil

CUI

CUI

Federal Agency: Army

Program Office: Army Contracting Command-Red River Army Depot (ACC-RRAD)

Aol Title: Track Pin and Bushing Manufacturing

CSO Aol Number: 004

Aol Release Date: 30 Oct 2025

Responses: See W911RQ-26-S-C001 for Procedure Details

General Information

Introduction

This announcement is for the design, manufacture, and installation of a tracked vehicle track pin and bushing manufacturing process Area of Interest (Aol) under the ACC-RRAD Commercial Solutions Opening W911RQ-26-S-C001. Interested parties are highly encouraged to visit the Government Point of Entry (www.sam.gov) to review the ACC-RRAD CSO special posting in its entirety to obtain more information on the process for submission and evaluation of the CSO Aol Responses.

This CSO Aol seeks to solicit innovative solutions that meet the objectives detailed in Section 1.2 and 1.3. The term “innovative” is defined as any technology, process, or method, including research and development that is new as of the date of proposal submission; or any new application of an existing technology, process or method as of the proposal date.

ACC-RRAD is not obligated to extend a request for proposal or make an award as a result of this CSO Aol; all awards are subject to the availability of funds and successful negotiations.

This CSO Aol is not an authorization to begin performance, and in no way obligates the Government for any costs incurred by the Offeror associated with developing a Response. Prior to commencement of any activities associated with performance under this CSO Aol, the Government will issue a written directive or contractual document signed by the Contracting Officer with appropriate consideration established. ACC-RRAD reserves the right to modify the requirements of this CSO Aol.

CUI

CUI

The authority to issue this CSO Aol is derived from ACC-RRAD Commercial Solutions Opening W911RQ-26-S-C001, dated 23 Sep 2025. The CSO authority is Section 803 of the National Defense Authorization Act for FY2022, Public Law 117-807, and Class Deviation 2022-O0007.

1.2 Aol Background

Red River Army Depot (RRAD) Rubber Products Division has a need for the design, manufacture, installation, and integration of a tracked military vehicle track pin and bushing manufacturing process (see Fig 1 Below).



Figure 1 – Track Pin and Bushing Examples

1.3 Fully Developed Track Pin and Bushing Manufacturing Process

RRAD is interested in a Commercial Solutions Offerings (CSO) for the design, manufacture, and installation of a tracked military vehicle track pin and bushing manufacturing process. We are looking for advanced manufacturing capabilities to develop a new process at RRAD to allow the capability of producing quality track pins and their bushings organically at this facility.

At a minimum RRAD is requesting that a new track pin/bushing process be able to produce pins and bushings to the specifications listed in the drawings.

General Metal Pin Fabrication Standards (May vary from pin to pin)

- Dimension range – Outer diameter of pins varying from approximately 1.1” to 1.5” and length measuring from approximately 12” to 28” depending on the pin type.

CUI

CUI

- Raw Material – Tubing or Bar Stock, Steel
- Production Space - Approximately 25,000 **DISCONTIGUOUS**, indoor square feet are available for production processes, and 10,000 outdoor, **DISCONTIGUOUS** square feet are available for possible storage. **If more space is needed, please specify and include in any provided cost estimates.**
- Quench and tempering
- Decarburization limits
- Shot-peening
- Induction hardening of pins
- Deburring
- Swage/Stamp/Press sections of pins
- Machine pins to specifications for internal diameter, outside diameter, taper, radii
- Room Temperature Pin Straightening
- Quality Inspections – Statistical process control Cpk standards
 - Dimensions
 - Flatness with surface plate, dial caliper indicator
 - Mag particle testing or Eddy Current inspection
 - Micrometer
 - microhardness
 - Parallelism with gage/dial indicator
 - Surface microfinish and/or texture with profilometer or comparator blocks
 - Decarburization – photo micrograph
 - X-ray diffractometer
 - Shot-peening
 - Cleanliness
 - Chemical analysis

General Rubber Bushing Fabrication Standards

- Mold rubber
- Vulcanize rubber to pin
- Quality Inspections-
 - Specific gravity
 - Hardness
 - Tensile strength
 - Ultimate elongation
 - Tensile stress
 - Thermogravimetric analysis

CUI

CUI

- Vulcanization using oscillation disk cure meter
- Adhesion
- Compression set
- Brittleness
- Gas Chromatography-Mass Spectrometry
- Bushing fill
- Concentricity
- Workmanship
- Product marking
- Endurance Bushing width and spacing
- Flash Thickness

Applicable Drawings, Standards, and Specifications:

T-161 (Approximate quantity of 170K pins per year)

- Drawing 12496791 (pin)
- Drawing 12496790 (pin with bushing)
- DOD-STD-00100D (AR) - Standard Practice for Engineering Drawing Practices
- ANSI Y14.5M-1982 - Dimensioning and Tolerancing
- MIL-DTL-11891 - Track Shoe Sets, Track Shoe Assemblies, Track Shoe Pads and Track Shoe Bushings, Vehicular: Elastomerized
- ASTM D2240 - Standard Test Method for Rubber Property—Durometer Hardness
- ASTM D395 - Standard Test Methods for Rubber Property—Compression Set
- ASTM D2137 - Standard Test Methods for Rubber Property—Brittleness Point of Flexible Polymers and Coated Fabrics
- TABLE III OF MIL-DTL-11891 - Track Shoe Sets, Track Shoe Assemblies, Track Shoe Pads and Track Shoe Bushings, Vehicular: Elastomerized
- Specific Gravity: ASTM D792 - Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement
- Hardness: ASTM D2240 - Standard Test Method for Rubber Property—Durometer Hardness
- Tensile Strength, Ultimate Elongation, Tensile Stress @200% Strain (Modules): ASTM D412 - Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers—Tension
- Tensile Strength (Aged): ASTM D573 - Standard Test Method for Rubber—Deterioration in an Air Oven
- Thermogravimetric Analysis Per ASTM E1131 - Standard Test Method for Compositional Analysis by Thermogravimetry

CUI

CUI

- Vulcanization Using Oscillating Disk Cure Meter: ASTM D2084 - Standard Test Method for Rubber Property—Vulcanization Using Oscillating Disk Cure Meter
- Adhesion: ASTM D429 - Standard Test Methods for Rubber Property—Adhesion to Rigid Substrates
- Compression Set: ASTM D395 - Standard Test Methods for Rubber Property—Compression Set
- Brittleness Point of Flexible Polymers: ASTM D2137 - Standard Test Methods for Rubber Property—Brittleness Point of Flexible Polymers and Coated Fabrics

T-107 (Approximate quantity of 70K pins per year)

- Drawing 8705896 (pin)
- Drawing 8705918 (pin with bushing)
- DOD-STD-00100D (AR) Engineering drawings
- ANS Y14.5M-1982 - Dimensioning and Tolerancing
- MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes
- MIL-S-13165 Shot Peening of Metal Parts
- ASTM-A304 Alloy Steel Bars Subject to End-Quench Hardenability Requirements
- ASTM-A519 Seamless Carbon and Alloy Steel Mechanical Tubing
- MIL-DTL-11891- Track Shoe Sets, Track Shoe Assemblies, Track Shoe Pads and Track Shoe Bushings, Vehicular: Elastomerized

T-158 (Approximate quantity of 60K pins per year)

- 12348381 Pin
- 12348382 Pin with bushing
- MIL-S-13165 - Military Specification: Shot Peening of Metal Parts
- ASTM E 384 Standard Test Method for Microhardness of Materials
- MIL-STD-271 Requirements for Non-destructive Testing Methods
- ASTM E1444 Standard Practice for Magnetic Particle Examination
- MIL-STD-1907 Inspection, Liquid Penetrant and Magnetic Particle, Soundness, Requirements for Materials, Parts, and Weldments
- AMS 6424 Steel Bars, Forgings, Mechanical Tubing and Forging Stock
- MIL-S-12515 Surface Hardening, Flame and Induction (For Ferrous Alloys)
- AMS 2301 Steel Cleanliness, Aircraft Quality Magnetic Particle Inspection Procedure
- MIL-DTL- 11891 - Track Shoe Sets, Track Shoe Assemblies, Track Shoe Pads and Track Shoe Bushings, Vehicular: Elastomerized
- Hardness: ASTM D2240 - Standard Test Method for Rubber Property—Durometer Hardness

CUI

CUI

- Tensile Strength, Elongation, Tensile Stress: ASTM D4 12 - Standard Test Methods for Vulcanized Rubber and Thermoplastic Elastomers—Tension
- Adhesion: ASTM D429 - Standard Test Methods for Rubber Property—Adhesion to Rigid Substrates
- ANSI Z1.1 Guide for Quality Control – Control Chart Method of Analyzing Data
- MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes
- ASTM D412 Test Methods for Rubber Properties in Tension
- ASTM D429 Standard Test Methods for Rubber Property—Adhesion to Rigid Substrates- Rubber Properties, Adhesion
- ASTM D792 Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement
- ASTM D2240 Standard Test Method for Rubber Property—Durometer Hardness - Rubber Hardness, Durometer
- ASTM D3182 Standard Practice for Rubber—Materials, Equipment, and Procedures for Mixing Standard Compounds and Preparing Standard Vulcanized Sheets - Rubber - Preparation of Standard Vulcanized Sheets

Aol Constraints

Any selected company/companies will be required to be registered in the System for Award Maintenance (sam.gov).

Submission Requirements

A written solution brief must be sent via e-mail submission by 5:00 P.M. (CDT) on 31 Dec 2025, with subject line stating “Track Pin and Bushing Manufacturing” to Jacob.j.horn.civ@army.mil and usarmy.rrad.acc.mbx.cso@army.mil.

Solution briefs received after the specified due date and time will be discarded and not be evaluated by the Government.

Aol Open Period – This Aol shall close at 5:00 P.M (CDT); 31 Dec 2025.

Point of contacts:

Aol questions can be sent to: Jacob.j.horn.civ@army.mil

Contracting Officer: christopher.l.johnston20.civ@army.mil

CSO Aol POCs:

usarmy.rrad.acc.mbx.cso@army.mil

CUI